

01 About the Team — Code Mates | Symbiosis Institute of Technology, Pune

Srishti Tripathi

B.Tech AI & ML | SIT Pune | CGPA 7.73

Python • Flask • LLMs • Multi-Agent AI • UI/UX

Key Projects: Quantum Advisor (multi-agent stock AI), Compliantly (ML classifier)

Achievements: Flipkart Grid 7.0 Semi-finalist, Bajaj HackRx 5.0 Semi-finalist, GSSoC 2025 Tech Contributor, 3rd place AI4Youth Ideathon

Utsav Raj Singh

B.Tech CSE | SIT Pune | 2023–2027

React • Node.js • Python • ML • Docker • REST APIs

Key Projects: AI Complaint Classifier (full-stack + Docker), Auto-Grader OCR System (Tesseract + TrOCR)

Achievements: Top 50 Bajaj HackRx 5.0 (800+ teams), SIH 2024 Round 1 cleared, 3rd place AI4Youth Ideathon

02 Problem Understanding

India's digital-payments boom masks a deeper crisis: millions have financial access but lack the knowledge to use it safely. The gap is sharpest for four underserved personas:

- **Priya (Thane, ₹35K/month teacher)** — overwhelmed by bank mis-selling, no trusted Marathi-speaking advisor
- **Rajesh (Mumbai gig worker, ₹15–40K variable)** — no emergency fund, prey to predatory loan apps
- **Kisan (Karnataka farmer, seasonal income)** — confused by PM-KISAN / KCC applications, no Kannada-language guidance
- **Divya (Pune, visually impaired content writer)** — banking apps not screen-reader friendly; financially dependent on family for basic tasks

Core insight: financial access ≠ financial understanding. Existing tools are English-first, salaried-professional-centric, and inaccessible to people with disabilities.

03 Proposed Solution — ArthSaathi

An Agentic, multilingual, voice-first AI companion built on a 5-agent architecture coordinated by a Supervisor Agent that routes every query based on the User Context Model built during onboarding.

System Architecture



The 5 Specialist Agents

Onboarding Agent	Conversational intake (voice/text) to build a living User Context Model — income type, language, life goals — no forms, no jargon.
Education Agent	Generates micro-lessons dynamically per user context: crop failure analogies for Kisan, gig-income tips for Rajesh, Marathi content for Priya.
Planning Agent	Variable-income budgeting splits, crop-cycle financial calendars, salary optimisation, education corpus modelling — context-aware, not generic.
Fraud Guard Agent	Real-time scam scoring of WhatsApp forwards and verbal descriptions. Cross-references live RBI/SEBI alerts with plain-language trust scores.
Accessibility Agent	Converts charts to spoken narratives. Full screen-reader support (WCAG 2.1). WhatsApp/USSD fallback. Offline content bundles for low-connectivity zones.

Key Differentiators

- ▶ **Saathi Score** — a single 0-100 wellness score visualised as a growing plant; gamification without intimidation
- ▶ **Community Circles** — NGO/educator dashboard to push curated content and track aggregate (anonymised) impact at scale
- ▶ **Responsible AI Guardrails** — vulnerability flag escalates distressed users to iCall / SEBI SCORES; AI never replaces licensed advice for high-stakes decisions
- ▶ **Outcome Metrics** — tracks real decisions (predatory loan avoidance, govt. scheme enrolment, savings consistency) not just app opens

04 Technology Stack — Exact Versions (Windows 11 Development Environment)

All packages listed below will be pre-installed. No downloads will be required on the event day.

Layer	Tool / Package	Version
OS Platform	Windows 11 Home / Pro (64-bit)	23H2
Runtime — Python	Python	3.11.9
Runtime — Node.js	Node.js (build tooling)	20.14.0 LTS
LLM API	Groq API — LLaMA 3.1 70B (free public tier)	openai-python 1.35.x
Agent Framework	Custom multi-agent orchestrator (Python)	—
Web Framework	Flask + Flask-CORS	3.0.3 / 4.0.1
Frontend	React.js + Tailwind CSS	18.3.1 / 3.4.4
Speech ASR	Bhashini API+ Whisper (local)	20231117
Speech TTS	gTTS (Google Text-to-Speech)	2.5.1
ML / NLP	Scikit-learn + NLTK + TF-IDF	1.5.0 / 3.8.1
Database	SQLite3 (local dev) + MongoDB	3.x / 7.0
Auth & Security	Flask-Login + bcrypt	0.6.3 / 4.1.3
Containerisation	Docker Desktop for Windows	27.0.3
HTTP Client	Axios (frontend)	1.7.2
Environment Mgmt	python-.env	1.0.1
Package Manager	pip + npm	24.x / 10.x

05 Impact Measurement — Outcomes, Not Vanity Metrics

ArthSaathi is built to prove real-world financial behaviour change, not just engagement numbers:

Scam Detection	Decision Quality	Scheme Enrolment	Savings Consistency
Flag predatory schemes & WhatsApp forwards in real-time before users act	Track whether users avoided bad loans / unnecessary charges this month	Measure uplift in govt. scheme (PM-KISAN, KCC, EPF) successful applications	Monitor emergency buffer-building for irregular & seasonal income users

Data is collected anonymously and in aggregate. Individual financial data stays on-device; the cloud receives only anonymised behavioural signals.

06 Responsible AI — Non-Negotiable Commitments

- ▶ **Transparency:** AI identity always disclosed; investment advice above regulatory threshold always referred to SEBI-registered advisors
- ▶ **Data Minimalism:** income ranges (not exact figures), goals (not bank statements); full profile deletion available at any time
- ▶ **Harm Prevention:** vulnerability detection flag escalates crisis signals to human helplines; system never auto-advises on financial distress patterns
- ▶ **Inclusive by Design:** WCAG 2.1 compliant, voice-first, USSD / WhatsApp fallback, offline content bundles for low-connectivity rural areas
- ▶ **Groq API Usage:** free public-tier LLaMA 3.1 70B via Groq — no proprietary lock-in, auditable open weights, rate-limit-aware fallback to local quantised model

"One AI companion — every language, every income, every ability."